Challenges and opportunities in quantum optimization

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Review article

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Challenges and opportunities in quantum optimization

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Abstract

Quantum computers have demonstrable ability to solve problems at a scale beyond brute-force classical simulation. Interest in quantum algorithms has developed in many areas, particularly in relation to mathematical optimization – a broad field with links to computer science and physics. In this Review, we aim to give an overview of quantum optimization. Provably exact, provably approximate and heuristic settings are first explained using computational complexity theory, and we highlight where quantum advantage is possible in each context. Then, we outline the core building blocks for quantum optimization algorithms, define prominent problem classes and identify key open questions that should be addressed to advance the field. We underscore the importance of benchmarking by proposing clear metrics alongside suitable optimization problems, for appropriate comparisons with classical optimization techniques, and discuss next steps to accelerate progress towards quantum advantage in optimization.

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Key points

- Quantum computing is advancing rapidly, and quantum optimization is a promising area of application.
- Quantum optimization algorithms — whether provably exact, provably approximate or heuristic — offer opportunities to demonstrate quantum advantage.
- Systematic benchmarking is crucial to guide research, track progress and further advance understanding of quantum optimization.
- Theoretical research and empirical research using real hardware can complement each other, in the move towards demonstrating quantum advantage.

Introduction

Quantum computing has the potential to revolutionize optimization and thereby impact science and numerous businesses. Optimization problems arise almost everywhere, and improvements over state-of-the-art classical algorithms using quantum computers could occur across multiple dimensions, such as solution quality, solution diversity, time-to-solution and cost-to-solution. But how tangible are the benefits of quantum optimization? Grover's search[1], quantum annealing, adiabatic quantum optimization[2,3] and the quantum approximate optimization algorithm (QAOA)[4] provided a promising start, but it has become clear that more is needed to realize a practical quantum advantage in optimization. Here, we address – for researchers and practitioners in quantum computing, optimization and related domains – the potential of quantum optimization from various angles: complexity theory, problem classes and algorithmic design, fair and robust benchmarking and the next steps towards real-world applications.

For some combinatorial optimization problems in a worst-case setting, the required effort using known classical approaches scales exponentially with the problem size[5]. Quantum computers may offer a quadratic speedup (for example, via Grover's search) – but a quadratic speedup over an exponential run time is still an exponential run time. The situation can, however, change drastically if we consider a concrete problem instance instead of the general worst-case scenario. Classically, many algorithms and heuristics can obtain an (almost) optimal solution in a reasonable time, even for some large problems. The most prominent example is the travelling salesperson problem (TSP), which is often used to illustrate the difficulty of combinatorial optimization but, in practice, very large problem instances can be solved close to optimality classically[6–11]. By contrast, there exist problems with fewer than 100 variables that are difficult to tackle classically[12,13]. Although quantum optimization algorithms will not necessarily improve performance for all instances of a problem, they do provide additional tools with new properties that can improve performance for some problem instances and thus improve our overall capabilities in optimization. It is crucial to understand exactly where advantages might lie, theoretically and empirically. Furthermore, synergies with classical algorithms can certainly improve the performance of quantum algorithms. We discuss these points formally, by laying the foundation for quantum optimization algorithms within various well-studied problem classes.

In practice, it is often necessary to rely on optimization heuristics, in which the performance for given problem instances is not known upfront. This makes comparing quantum and classical algorithms rather tricky, given the limitations of what can be run on today's quantum hardware. To establish systematic, reproducible and comparable benchmarks, we introduce fair metrics and relevant optimization problems to which different algorithms can be applied. This is crucial to identify optimal strategies and guide the development of new quantum optimization algorithms. Finally, we propose next steps towards quantum advantage.

Quantum advantage and complexity theory

In theoretical computer science, complexity classes group computational problems together in terms of the amount of resources, such as computational time and memory, required to solve them. Complexity theory is thus often the guiding compass when seeking classes of problems that are hard for classical computers to solve, but perhaps demand less resource on quantum computers. This is a reasonable and rigorous way to formulate a quantum advantage, but proving that problems require strictly more resources classically than they would on a quantum computer can be a monumental task. Additionally, complexity theory is traditionally concerned with worst-case instances of a problem: in practice, the concrete problem instance to be solved might be easier than the hardest, worst-case instances[14,15]. Hence, even when worst-case complexity-theoretic results rule out quantum speedups for hard problems, quantum computers might still outperform classical computers for certain classes of problem instance. Our notion of quantum advantage in practice should cast the net wider than formal proofs and worst-case scenarios, but within a reasonable range guided by complexity-theoretic arguments.

Categorizing the complexity of an optimization problem depends on its formulation and the desired quality of the solution. Much of the complexity theory deals with decision problems, that is, deciding whether a particular solution to a problem exists or not. The most notable complexity classes containing decision problems are labelled P (polynomial time) and NP (non-deterministic polynomial time): P contains the set of decision problems known to require an amount of time to solve that grows at most polynomially with the input size; NP contains problems whose solutions are verifiable in polynomial time. In practice, finding an actual solution to a problem is more important than determining its existence. For example, in the TSP, merely knowing that there exists a path of a certain length to travel to all cities and back is not enough to benefit; the actual path to travel is needed. This leads to the notion of relational problems, which generalize decision problems and are classified by their own functional complexity classes, denoted FP and FNP. The classes PO $\subset$ FP and NPO $\subset$ FNP are the sets of optimization problems analogous to P and NP (ref. 16). (A comprehensive overview of various complexity classes pertinent for practical scenarios is given in the Supplementary information.)

Although it is widely believed that quantum computers will not offer exponential speedups for NPO-hard problems (the most difficult optimization problems to solve), the story does not end there. An interesting open question is whether quantum algorithms can solve optimization problems that are average-case hard for classical algorithms. In particular, problems with a so-called overlap gap property[17] have been proven to be average-case hard for certain families of classical algorithms (and local low-depth quantum algorithms), but it remains unclear whether these problems can be addressed by higher-depth or more sophisticated quantum algorithms[18–22]. If so, this would be very convincing evidence for useful superpolynomial quantum advantage, given that such problems naturally arise in combinatorial optimization, statistical physics and high-dimensional statistics.

Additionally, in practice, an exact solution to a problem might not be required. An approximation to the exact solution is often acceptable, leaving room for the possibility of exponential quantum speedups in realistic settings. Studying the ability of quantum algorithms to approximate solutions – that is, the hardness of approximation with quantum computing – raises many questions. For example, there are empirical investigations into whether quantum algorithms can achieve better approximation ratios (a measure of how close an approximate solution is to optimality) than classical methods[23–26].

An 'approximable' problem is said to belong to the complexity class APX, which contains the set of NPO problems for which there are polynomial-time approximation algorithms with approximation ratios bounded above by some constant c (ref. 27). An interesting subclass of NPO problems that admit a so-called polynomial-time approximation scheme is PTAS (refs. 15,16). For any problem in PTAS, there is a polynomial-time algorithm that is guaranteed to find a solution whose value is within a factor $1+\varepsilon$ of the optimum, in which $\varepsilon > 0$. However, there are cases in which the exponent of the polynomial might even depend exponentially on $1/\varepsilon$. To account for this, the class FPTAS – for fully polynomial-time approximation scheme – can be considered, which demands a polynomial run time in the input size and in $1/\varepsilon$. The hierarchy of these optimization problem classes is shown in Fig. 1, and the hardness of approximation is discussed more formally in the Supplementary information. When combined with quantum computing, this area makes for an exciting and rather unexplored research direction – where quantum algorithms could serve as entirely new approaches or improve over existing classical PTAS algorithms.

Relaxing the requirement of a provably approximate solution to one deemed 'good enough' (by some reasonable standard) enables heuristic approaches, which are arguably of greatest practical relevance. Heuristic algorithms are understood as algorithms without provable performance or running time guarantees. The success of classical heuristic algorithms motivates the study of quantum heuristics. Here, the aim is to design algorithms that leverage quantum computation to provide reasonable solutions to problems where classical approaches either cannot or take too long[15]. The word 'reasonable' is context-dependent, making heuristic approaches difficult to analyse theoretically. Noteworthy efforts are discussed in the Supplementary information.

Although substantial progress has been made in the quantum and classical complexity theory[28–30], the discussion mentioned earlier highlights issues that arise when restricting our notion of quantum advantage strictly to complexity-theoretic separations in the context of optimization. The key takeaway is that understanding complexity theory is extremely useful for gauging a possible quantum advantage in optimization, but rigorous complexity-theoretic separations are not necessary nor sufficient when seeking a practical quantum advantage.

Again, the TSP is an illuminating example. Its most general decision version is NP-hard and the optimization version is APX-hard. If the problem is restricted to the Euclidean TSP (in which all points are in the d-dimensional space $\mathbb{R}^d$ and edge weights equal their Euclidean distances), then, remarkably, there exists an explicit PTAS to solve such TSP instances[31]. Using heuristic algorithms, it is possible in practice to solve TSP problems with millions of nodes. Finally, if we instead consider the shortest-path problem (finding the shortest path between two places), we have a problem in P. This illustrates the nuance of complexity theory that should be kept in mind when designing quantum optimization algorithms. Importantly, if provable quantum speedups seem unattainable, quantum heuristic algorithms might still offer meaningful results. But developing quantum heuristic methods that exploit structure in a problem, or enhance classical approaches, is not straightforward: in Box 1, we lay out various building blocks for the careful design of quantum optimization algorithms.

Problem classes and algorithms

Optimization problems can be classified according to, for example, types of variables, objective function (functions), constraints and uncertainty. A problem can often be modelled in different ways. Although these formulations might be equivalent, the choice of formulation can have a crucial role in how a problem is solved. Indeed, different formulations place more or less emphasis on certain properties of the problem. Furthermore, the available algorithms might only work for certain problem classes. In addition, it is crucial to understand how the performance of algorithms changes when executed on noisy quantum hardware and how instances can be optimally mapped to the hardware to minimize noise. Because these questions are difficult to answer, systematic benchmarking is very important. Here, we focus on the most prominent problem classes in the literature and discuss available algorithms, based on the paradigms mentioned in Box 1. (More details and additional problem classes, such as robust optimization or multi-objective optimization, are given in the Supplementary information).

Discrete optimization

Unconstrained discrete optimization. Quadratic unconstrained binary optimization (QUBO) is a foundational example of unconstrained discrete optimization:

$$\min_{x\in\{0,1\}^n} x^T Qx. \tag{1}$$

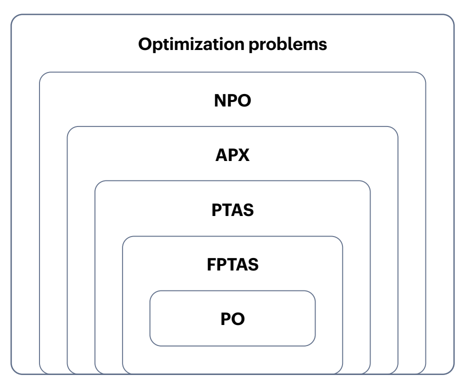

Fig. 1 | Complexity classes for optimization problems. In the hierarchy of complexity classes, NPO (which stands for non-deterministic polynomial-time optimization) is the set of optimization problems that is analogous to the NP class of decision problems, whose solutions are verifiable in polynomial time. An 'approximable' problem – for which there are polynomial-time approximation algorithms with guaranteed approximation ratios – belongs to the class APX (for approximable), within NPO and PTAS is a further subset, for which polynomial-time algorithms exist that are guaranteed to find a solution that is within a predefined factor $1\pm\varepsilon$ of the optimum, in which $\varepsilon > 0$ denotes the optimality target. FPTAS is a more restricted class than PTAS, containing problems that admit algorithms whose running time is polynomial in both input size and the inverse of the desired optimality target ε. Finally, PO (for polynomial-time optimization) is the class of optimization problems that is analogous to the polynomial-time class P. FPTAS, fully polynomial-time approximation scheme; PTAS, polynomial time approximation scheme.

Box 1 | Paradigms in the design of quantum optimization algorithms

Multiple approaches lead to exact quantum optimization algorithms, such as Grover (adaptive) search[225–227] and the quantum adiabatic algorithm (QAA). Here, we briefly introduce the core concepts deployed as subroutines in various quantum optimization algorithms. Many of them are expected to require fault-tolerant quantum computing. However, they also serve as theoretical motivation for approximation algorithms and heuristics, which may already lead to quantum advantages with noisy quantum computers. (More details can be found in the Supplementary information).

Grover search. The seminal Grover algorithm for searching an unstructured database achieves a quadratic advantage over classical search in terms of reducing the number of queries to the database[1]. In exact quantum optimization, this advantage extends to function minimization[77,225], achieving a quadratic speedup over brute-force search in combinatorial optimization problems. Although, in general, this is the best that can be hoped for[228], the advantage is often less pronounced in practical problem classes in which brute-force search is not the best classical algorithm. Still, combining Grover Search with other quantum or classical algorithms can lead to an overall quadratic speedup in specific cases[209].

Quantum adiabatic algorithm (QAA). QAA (refs. 2,3,44) is another exact optimization algorithm related to homotopy methods in numerical mathematics[229]. It relies on a problem Hamiltonian whose ground state represents the solution and a mixing Hamiltonian with an easy-to-prepare ground state. QAA gradually evolves the state by interpolating between these Hamiltonians, ensuring that the final state is close to the solution of the considered problem when the evolution is sufficiently slow. Although theoretically powerful, the — in general — exponential time required for precise adiabatic evolution limits its practicality as an exact optimization algorithm. However, QAA may be an effective heuristic in non-adiabatic implementations[44,168,169,174,230–233].

Quantum phase estimation (QPE). QPE estimates eigenphases of unitary operators[234,235], enabling the calculation of eigenvalues for specific matrices and the solving of Hamiltonian-related problems[236–238]. Shor's algorithm[239] leverages QPE to achieve an exponential speedup over the best-known classical algorithms for factoring and related problems[222,223]. QPE is also valuable for optimization when a problem is encoded into non-diagonal Hamiltonians[240,241].

Gibbs sampling. Gibbs sampling generates sample sequences from complex probability distributions, for example, by using a Markov-chain Monte Carlo algorithm[242]. Its significance in optimization, particularly in classical Markov-chain Monte Carlo algorithms, extends to solving problems such as constraint satisfaction[243]. Believed to be at least QMA-hard[244,245] (in which QMA stands for Quantum Merlin–Arthur, and is a complexity class, containing P and NP, of problems that are in general not believed to be efficiently solvable by quantum computers), quantum Gibbs sampling[246] has a crucial role in various optimization algorithms, such as discrete and convex optimization. Quantum speedups are possible because, under certain conditions, Gibbs states can be prepared more efficiently quantumly than classically[69,246–252].

Approximation algorithms. In addition to exact algorithms, there exist approximation algorithms that guarantee a certain approximation ratio. Quantum computing may accelerate known classical approximation algorithms via quantum subroutines. For instance, semidefinite programming relaxations — as used in the famous Goemans-Williamson algorithm for the maximum-cut problem[64,253] or in more general variants for the quadratic unconstrained binary optimization problem[254] — may profit from quantum semidefinite-programming solvers.

Variational quantum algorithms (VQAs). VQAs are a family of search algorithms, originally proposed for quantum chemistry applications[255], which utilize parameterized quantum states and cost functions to iteratively optimize parameters and find good solutions[4,256–258]. Although finding optimal parameters can be NP-hard[259,260], VQAs provide a heuristic approach with potential quantum advantages.

Here, x denotes the decision variables, $Q \in \mathbb{R}^{n\times n}$ is the cost matrix and $n \in \mathbb{N}$ is the number of decision variables. Other types of unconstrained discrete-optimization problems include higher-order polynomial objective functions[32] (polynomial unconstrained binary optimization, black-box objective functions[33] and (bounded) integer variables). Some problems, such as the maximum-cut (MAXCUT) problem, are naturally formulated as a QUBO, and most unconstrained discrete-optimization problems can be mapped to QUBOs[34–37], for example, by replacing integer variables with a suitable encoding[38,39] or by converting higher-order polynomial terms to additional binary variables and quadratic terms[40,41]. In general, QUBO is NP-hard[42]. QUBO is also APX-hard, that is, there cannot exist a PTAS for QUBO unless P = NP (ref. 43). In principle, Grover search (Box 1) or quantum annealing could be used to solve QUBOs; however, these algorithms with provable approximation-ratio guarantees are expected to require FTQC.

The relationship between solving a QUBO and finding the ground state of a Hamiltonian is fundamental to many quantum optimization algorithms. The most common translation first substitutes binary variables $x \in \{0, 1\}$ with spin variables $z = 1 - 2x$, in which $z \in \{-1, +1\}$ is known as the Ising model. These spin variables are then replaced with Pauli Z matrices, resulting in a diagonal Hamiltonian $H \in \mathbb{R}^{2^n \times 2^n}$ that encodes the objective values of equation (1). The ground state of H represents the QUBO's optimal solution[34].

The first quantum-computing approach proposed for solving unconstrained discrete-optimization problems was the quantum annealing algorithm[2,3,44]. Quantum annealing is a quantum adiabatic algorithm method that specializes in finding the ground state of a classical Ising model by conducting a continuous time evolution ('annealing') between an initial uniform superposition state and a target Ising Hamiltonian (Box 1). The run time of the quantum annealing algorithm is determined by the speed of the time evolution. The adiabatic theorem states that if the evolution is slow enough, this process will identify optimal solutions of the classical combinatorial optimization problem[45]. The amount of time needed to ensure that quantum annealing produces optimal solutions is proportional to the minimum gap in the eigenspectrum of the quantum Hamiltonian during the evolution process, which is known to shrink exponentially with n for many interesting optimization problems[46]. Given that quantum annealing can encode APX-hard computations, this result is an expected consequence of computational complexity theory. The continuous-time nature of the quantum annealing algorithm makes it difficult to analyse using purely mathematical methods. Hence, quantum annealing research tends to focus on empirical studies of the behaviour of the algorithm in ideal[47] and noisy computational settings[48,49].

The QAOA[4] is a variational quantum algorithm motivated by quantum annealing that is tailored for implementation as low-depth quantum circuits. The QAOA uses a particular problem-dependent ansatz given by pairs of unitaries in each layer. These unitaries are

$$U_X(\beta) = e^{-i\beta H_X}, \quad (2)$$

$$U_P(\gamma) = e^{-i\gamma H}, \quad (3)$$

and they are known as mixing and problem unitaries, respectively. Here, H denotes the problem Hamiltonian and H_X denotes the mixing Hamiltonian (see Supplementary information). Both unitaries are alternated and repeated p times, with individual parameters β_j, γ_j for $j = 1, \ldots, p$. The p layers are applied to the initial state $|+\rangle^{\otimes n}$.

In terms of positive results, there are certain problems and algorithmic settings in which the QAOA can be seen as an approximation algorithm with worst-case performance bounds. For example, there exists worst-case performance guarantees for QAOA with $p = 1, 2, 3$ layers for the MAXCUT problem on three-regular graphs. Specifically, for $p = 1$, a lower bound on the approximation ratio of 0.692 has been derived[4]; also a lower bound of 0.7559 for $p = 2$ and (under certain assumptions) 0.7924 for $p = 3$ (ref. 50). Aside from these and a handful of other limited settings, performance guarantees for the QAOA are in general not known and the QAOA thus remains a heuristic. Considering that the energy of the QAOA state for any Ising Hamiltonian can be efficiently and explicitly computed classically for $p = 1$ (ref. 51), this does not immediately imply a guarantee on the approximation ratio. The performance of the QAOA for (large-girth) d-regular graphs has been analysed numerically, and it can outperform many known relaxations based on semidefinite programming (SDP) from $p \geq 11$ (refs. 52,53).

Some variants adjust the QAOA to warm-start it from solutions obtained by classical algorithms, for example, by leveraging classical SDP relaxations and adjusting the QAOA mixers[54,55]. These approaches inherit some of the performance guarantees of the classical algorithms used for warm-starting and can greatly outperform the standard QAOA in certain instances. Other QAOA variants restart the search on encountering a barren plateau[56], that is, exponentially decaying gradients: the more the barren regions and the higher their proportion in the parameter space, the higher the speedup obtained by restarting.

Another alternative to the QAOA is the recursive QAOA (ref. 57), which uses a quantum computer to produce a sequence of reduced problem instances. Correlations between variables are estimated iteratively, and the problem Hamiltonian is reduced by identifying the strongest pair-wise correlation and replacing one of the involved variables by ±1 times the other variable, in which the sign depends on the sign of the correlation. This process is iterated until the reduced problem becomes small enough for (provably exact or heuristic) classical solvers, which then yields an approximate solution to the input problem. Recent works[58–61] have generalized recursive quantum approaches to problems beyond MAXCUT and to circuit ansatz beyond the QAOA. Another family of near-term compatible approximation algorithms for MAXCUT is given by quantum random access optimization[62,63].

In terms of negative results, in the large-problem limit, a constant-depth QAOA cannot outperform the classical Goemans–Williamson approximation[64] for certain instances of MAXCUT on d-regular graphs – the depth needs to grow with the size of the problem[57]. However, these results are not applicable to alternative approaches such as other QAOA variants or the recursive QAOA. Further results on the limitations of the QAOA include that QAOA 'needs to see the whole graph' for a particular QAOA approach to the maximum-independent-set problem[65]; similar obstructions to the low-depth QAOA have also been shown to apply to k-local generalizations of MAXCUT ($k \in \mathbb{N}$) and constraint satisfaction problems in the average case[66,67].

A heuristic based on time evolution is quantum imaginary time evolution (QITE)[68,69]. Similar to the quantum adiabatic algorithm, QITE assumes an initial state $|\psi_0\rangle$ with a non-zero overlap with the ground state of H. Then, it applies the evolution

$$|\psi_\tau\rangle = \frac{e^{-H\tau}}{Z_\tau}|\psi_0\rangle, \quad (4)$$

in which $\tau = it$ (imaginary time) and $Z_\tau = \sqrt{\langle\psi_0|e^{-2H\tau}|\psi_0\rangle}$ denotes a normalization factor related to the partition function. QITE amplifies the amplitudes of the ground state and exponentially suppresses the amplitudes of the other eigenstates. However, because QITE is a non-unitary evolution, it is challenging to implement on a quantum computer[69], and only heuristic variants of QITE exist[33,69–72] at present. QITE would, however, make it possible to solve QMA-hard problems such as generic ground-state search or Gibbs state preparation[73].

Alternative cost functions can help to increase the robustness against noise or relax the requirements on an ansatz. For example, the conditional value at risk[74,75] does not average over the objective values corresponding to all samples obtained from measuring a quantum state, but sorts them and only takes the average over the best α-fraction of samples, for a predefined $\alpha \in [0, 1]$. Another cost function is the Gibbs objective function[76]. As with conditional value at risk, it biases the estimate by emphasizing good samples over bad samples through applying different weights.

Constrained discrete optimization. Constrained discrete optimization is defined by adding constraints of the type $g(x) = 0$ or $g(x) \leq 0$ to an unconstrained discrete-optimization problem. For simplicity, we consider QUBO and add linear inequality and equality constraints, although some of the algorithms are applicable to more complex types of constraints. The conversions of higher-order polynomial objective functions to quadratic functions and from integer to binary variables still apply as already discussed for unconstrained discrete optimization. Thus, the considered problems are

$$\begin{aligned} \min_{x \in \{0,1\}^n} \quad & x^{\mathrm{T}} Q x \\ \text{s.t.} \quad & Ax = \mathbf{b} \\ & Cx \leq \mathbf{d} \end{aligned} \quad (5)$$

for matrices A and C and vectors $\mathbf{b}$ and $\mathbf{d}$ of appropriate dimensions. Unlike for QUBOs, linear objective functions (that is, a diagonal Q) are interesting owing to the presence of constraints.

Under mild assumptions for A and C, equation (5) can always be converted to QUBO by adding slack variables to convert all inequality constraints to equality constraints and then converting equality constraints to penalty terms by squaring the right-hand side and adding it to the objective function with a large weight[36]. Some constraints can also be encoded into penalty terms more efficiently (see, for example, ref. 36); however, although always possible, this might come at a substantial cost in terms of additional binary variables and a dramatic increase in non-zero terms in the cost matrix in equation (5) – that is, it often maps an otherwise sparse matrix Q to a dense one.

We will now discuss strategies to incorporate constraints directly. As in unconstrained discrete optimization, Grover search can be leveraged to achieve a quadratic speedup over brute-force search[77]. This requires not only an oracle for the objective function but also one for each constraint, to mark all feasible solutions. Similarly, a quantum branch and bound algorithm can accelerate classical branch-and-bound algorithms[78]. The algorithm adapts Grover search to search branch-and-bound trees, but has the disadvantage that no a posteriori bounds on the optimality gap are generated, which often enable an early termination of such search schemes.

There are multiple proposals to incorporate constraints in variational quantum algorithms. The quantum alternating operator ansatz[79,80] generalizes the QAOA to much wider classes of problems and encodings, in particular to problems with hard constraints. Here, more general initial states are considered, such as ones from the subspace of feasible states for constrained problems. Alternatively, one can also directly replace the mixing Hamiltonian H_X in equation (2), for example, for Hamming-weight constraints by an XY-model Hamiltonian[81,82], or by a Grover-based[83,84] Hamiltonian. However, the resulting mixers are more complex and challenging to implement. Another approach is to leverage quantum Zeno dynamics to repeatedly project the state back to the feasible subspace[85]. Or iterative or recursive QAOA variants may be adapted to enforce constraints as suggested in ref. 60. Here, as for the unconstrained case, problems are solved iteratively and usually one variable is removed at a time. Adjusting the selection rules for how variables are fixed makes it possible to enforce constraints in certain cases.

Continuous optimization

Convex optimization. A quite general convex optimization problem is to minimize a convex function $f:\mathbb{R}^n\to\mathbb{R}$ over all $x\in\{x\in\mathbb{R}^n:\|x\|_2\le R\}$ for a norm upper bound $R>0$, given black-box access to evaluate f and possibly its derivatives. Gradient descent, for example, works in this framework.

A natural question is whether quantum computers offer a speedup in query complexity or run time, and the answer is not always positive. For example, refs. 86,87 consider black-box access to f and its (sub) gradients. In this setting, classical and quantum algorithms have the same lower bound for the query complexity in general: no n-dependence, but a polynomial scaling in the inverse of the desired precision ε. Alternatively, a polynomial n-dependence can be allowed, but only a polylogarithmic dependence on ε. Then, under mild assumptions, general convex optimization problems can be solved in polynomial time using, for example, the ellipsoid method[88]. In this general setting, the aim can be to solve a problem of the form $\min c^T x$ s.t. $x\in K$, for some bounded convex set K, to which there are various types of oracle access (membership, separation and optimization). Here mild quantum speedups (and no-go results) are known[89,90].

Often, much more can be said when the convex optimization problems have a certain structure. The most notable such classes are arguably linear programming and SDP. Here, the objective is to minimize a linear function subject to linear inequalities on non-negative vectors (for linear programming) or on positive semidefinite matrices (for SDP). SDP can be expressed in the form

$$\begin{aligned} \max_{X\succcurlyeq 0}\ & \operatorname{tr}(CX) \\ \text{s.t.}\ & \operatorname{tr}(A_iX)\le b_i\ \ \forall\, i\in[m], \end{aligned} \tag{6}$$

in which the problem is defined by symmetric matrices $C, A_1, \ldots, A_m\in\mathbb{R}^{n\times n}$ and a vector $b\in\mathbb{R}^m$. Linear programming is a special case of SDP in which each of the input matrices is diagonal. Quantum algorithms for these problems usually trade off an improved dependence on the dimension n and the number of inequalities m with a worse dependence on the precision ε with which the problem is solved. Quantum algorithms for these problems can be categorized based on the solution methodology.

The first quantum algorithms for SDP were first-order methods, based on the multiplicative weights update (MWU) framework[91]. In this framework, candidate solutions X are proportional to $\rho=\exp(\sum_{i=1}^m y_iA_i)/\operatorname{tr}(\exp(\sum_{i=1}^m y_iA_i))$ for some (sparse) vector $\mathbf{y}\in\mathbb{R}^m$ – that is, proportional to Gibbs states, which naturally correspond to trace-normalized positive semidefinite matrices and can be efficiently prepared on a quantum computer under some conditions. To implement the MWU framework, it is also necessary to compute trace inner products $\operatorname{tr}(A\rho)$ between such Gibbs states ρ and a matrix A (which is either one of the A_i's or C) and essentially solve a search problem based on these values.

All of these steps can be performed on a quantum computer, with various levels of efficiency. The run time of quantum MWU methods depends on several parameters: the dimension n, the number of constraints m and the desired (additive) error ε in the objective function and on two instance-specific parameters, typically denoted by R, and r, that are related to the diameter of the feasible region for the primal ($\operatorname{tr}(X)\le R$) and dual problems ($\|y\|_1\le r$) and bound the so-called width of the oracle. Initial results had a run time of $\tilde{\mathcal{O}}(\sqrt{mn}\,s^2\operatorname{poly}(\frac{Rr}{\varepsilon}))$, in which s is the row sparsity of the input matrices and $\tilde{\mathcal{O}}$ ignores polylogarithmic factors in all variables[92,93]. Later work[94–96] has improved this to $\tilde{\mathcal{O}}((\sqrt{m}+\sqrt{n})s^2\operatorname{poly}(\frac{Rr}{\varepsilon}))$.

In comparison with classical algorithms, the best-known general SDP solvers, which approximate the optimal value to additive error ε, scale as $\tilde{\mathcal{O}}(m(m^2+n^\omega+mns))$, in which $\omega\in[2, 2.373)$ is the unknown optimal matrix exponent for multiplication[97]. Thus, quantum SDP solvers seem to be beneficial in the regimes where n and m are large; however, the input models of the classical and quantum algorithms are not necessarily equivalent so it is not entirely fair to compare these results. Furthermore, the parameters r, R and ε usually appear together as one 'scale-invariant' parameter $\gamma=Rr/\varepsilon$, which can scale poorly in terms of n and m (ref. 93), especially when r and R scale linearly or superlinearly with n (see Supplementary information for more details). Notable exceptions are applications to shadow tomography, quantum state discrimination and E-optimal design (optimal design in which the smallest eigenvalue of the information matrix is maximized)[94,95]. First-order methods other than MWU could also be studied (see, for example, ref. 98), but the speedup there is poorly understood.

Interior point methods (IPMs) are popular second-order methods to solve structured convex optimization problems both from a theoretical and from a practical perspective. IPMs start from a point lying in the interior of a convex feasible set $\mathcal{X} \subset \mathbb{R}^n$ and use Newton's method to solve a sequence of barrier problems:

$$\min_{x \in \mathbb{R}^n} \eta \cdot \langle c, x \rangle + f(x), \tag{7}$$

in which $\langle \cdot, \cdot \rangle$ is an inner product on $\mathbb{R}^n$, $\eta \geq 0$ and f: $\mathrm{int}\,(\mathcal{X}) \mapsto \mathbb{R}$ is a barrier function encoding the constraints defining $\mathcal{X}$. That is, $f(x) \to \infty$ as x approaches the boundary of $\mathcal{X}$, and thus the only 'constraint' present in equation (7) is that f is only defined on the interior points of $\mathcal{X}$. Each step requires solving a linear system. It is therefore natural to ask whether quantum linear-system solvers[99–101], followed by quantum state tomography, can be used to speedup such a task. However, multiple issues emerge. The use of state tomography introduces inexactness in the search direction and a $1/\varepsilon$ dependence in the run time, in which ε is the tomographic precision[102]. Furthermore, using the quantum linear-systems algorithm introduces a run time dependence on the condition number of the Newton linear system, which, unfortunately, goes to ∞ as optimality is approached — although these two issues can be partly dealt with by IPMs that allow inexact search directions[103] and by iterative refinement techniques[104,105]. Overall, assuming access to quantum random access memory (QRAM), quantum IPMs achieve favourable dependence on the size of the problem (m and n) for linear and semidefinite optimization compared with classical algorithms, but the remaining dependence on some numerical parameters (such as a condition number bound for Newton systems) makes it unclear whether an overall speedup is achieved. Although concrete proposals to build QRAM exist[106,107], a scalable fault-tolerant implementation is yet to be found[108–110], making this at best a long-term prospect.

Only very recently, a quantum speedup for IPMs was achieved without introducing a dependence on a condition number[111]. Under some mild assumptions and access to QRAM, this IPM achieves a quantum speedup for linear programmes in which the number of linear inequalities m is much larger than the number of variables n. In particular, the best classical solver achieves a scaling of $\tilde{\mathcal{O}}(mn + n^{2.5})$[112], whereas the quantum analogue scales as $\sqrt{m} \cdot \mathrm{poly}(n, \log(1/\varepsilon))$.

Quantum algorithms for convex optimization can also be designed by providing a Schrödinger equation whose trajectory follows a descent path (as in gradient descent) and simulating the corresponding dynamical system. This approach was pioneered in ref. 113, leading to a quantum algorithm with convergence guarantees on convex and non-convex problems, although the latter may require exponential time in general. The approach has been extended[114] to follow the central path for linear programming, giving an alternative approach to IPMs with better running time in most parameters, even without QRAM.

Non-convex optimization. In general, non-convex optimization can be defined as:

$$\min_{x \in \mathbb{R}^n} f(x). \tag{8}$$

Many variants are undecidable[115], especially in the cases of periodic functions f, or sufficiently non-smooth functions, but kth order critical points of non-convex functions can be studied.

There are clear limits to the speedup in non-convex optimization[116], echoing the situation in convex optimization[86]. In particular, when the first $k \in \mathbb{N}$ partial derivatives of f are available via an oracle or there is access to stochastic gradients, and no further assumptions are made, classical lower bounds apply to quantum algorithms too. Hence, suitable special cases are sought, such as by considering the numbers of local minima, local behaviour of the functions around those minima and behaviour of the trajectories connecting nearby local minima (the 'barriers').

It has also been shown[117] that there are query upper bounds on finding critical points with quantum algorithms for stochastic gradient descent. With access to a probability-weighted superposition over the bounded-variance stochastic gradient, the algorithm requires $O(\sqrt{n}\,\varepsilon^{-3})$ queries, which is only better than the classical algorithm $O(\varepsilon^{-4})$ when the dimension is lower than ε^{-2}. This can be improved with access to mean-squared smooth stochastic gradients, from $O(\varepsilon^{-3})$ to $O(\sqrt{n}\,\varepsilon^{-5/2})$. The run time for all of these algorithms both classically and quantumly will, in most cases, be polynomial in n.

In high-dimensional situations, the availability of the quantum linear-systems algorithm and of quantum linear-algebra methods motivates the direct quantum implementation of gradient descent over $O(\mathrm{polylog}(n))$ qubits[118,119] to provide descriptions of stationary points of non-convex functions. With a non-convex differentiable function $f(x): \mathbb{R}^n \to \mathbb{R}$ and an initial point $x^0 \in \mathbb{R}^n$, consider the gradient descent update $x \to x - \eta \nabla f(x)$ with $\eta > 0$. For $T \in \mathbb{N}$, the task is to output a quantum state proportional to x^T, in which the update is applied T times starting from x^0. Although the algorithm may admit a polylog(n) complexity, amplifying the $|0\rangle$ state at the end of the gradient procedure can cost $O(2^T)$ and an efficient circuit may be constructed only for special cases[118,120] or could be approximated via pre-training a parameterized quantum circuit.

Mixed-integer programming

Combining discrete decision variables and continuous decision variables, mixed-integer programmes are non-convex and at least as hard as the discrete-optimization and continuous-optimization problems they subsume.

There are several potential algorithms to approach mixed-integer programming (MIP) using a quantum computer. 'Branch-and-bound-and-cut' is the main workhorse in classical MIP. One possibility for a hybrid implementation is that discrete decisions are handled by branching and generation of cuts implemented classically, whereas the continuous, possibly convex relaxations could be solved on the quantum computer. Considering the progress in quantum algorithms for convex optimization, this approach seems plausible, albeit distant. More ambitiously, the branching decisions could be determined quantumly as well, which would yield an additional quadratic speedup[78,121–123] over the speedup of quantum algorithms for convex optimization, while increasing the requirements on the numbers of qubits and fault tolerance substantially. Some earlier works[78,121] consider the speedup in the case of locating all optima (measured against the classical cost of finding all optima), whereas others[122,123] consider the speedup of obtaining one optimum (out of possibly multiple ones), measured against the classical cost of finding one optimum. It has recently been suggested[124] that super-quadratic speedups might be possible in some cases.

Similarly, decomposition algorithms (such as splitting, or the alternating direction method of multipliers) for MIP often solve the QUBO subproblem using a quantum computer[125,126], while solving the continuous subproblem classically. Under some restrictive conditions[126], such an iterative approach can be shown to be globally convergent. A similar approach has been tested[127] in the context of Benders decomposition.

Alternatively, MIP can be reformulated to an unconstrained optimization problem. A decomposition has been explored[128,129] that minimizes values of the slack variables used to convert inequality constraints to equality constraints, with a Lagrangian relaxation of the resulting equality-constrained continuous-valued problem. Such Lagrangian methods are well understood[130,131], but global-convergence results are limited by very restrictive conditions, in general, or require some form of backtracking[132].

MIP could also be reformulated to high-dimensional linear programming using techniques that date back to the work of Dantzig and Wolfe in the 1960s (ref. 133). Similarly, using moment or sum-of-squares techniques[134,135], MIP can be reformulated to infinite-dimensional linear programming, with an SDP relaxation in a dimension much higher than $n+d$ and sometimes[136] exponential in $n+d$. If quantum algorithms for SDP could make it possible to solve such instances with run time independent of the dimension, this might be an interesting avenue for quantum algorithms for MIP. Finally, a mathematically elegant reformulation of an MIP problem results in a linear programme over the dual of the cone of copositive matrices[137]. For a fixed n, these matrices form a convex cone[138], but even the test of membership in this cone is NP-hard[139]. A possible way to exploit this reformulation in the context of a hybrid solver[140] is to use a quantum computer to solve the difficult cut generation problem, whereas the classical computer solves a convex problem.

Dynamic programming

Dynamic programming is a very generic mathematical optimization method that breaks down large optimization problems into smaller ones and combines results for the overall problem in a recursive fashion. A dynamic programming algorithm consists of repeatedly optimizing the Bellman equation:

$$V(x) = \min_a T(x,a) + V(x_a), \tag{9}$$

in which a are the actions that are available in the state x and $x_a = F(x,a)$ is the state reached by applying a in the state x. The entries $V(x)$ can be arranged into a dynamic programming table. A dynamic programming algorithm then computes all the entries $V(x)$ in this table. The order in which the entries are computed is chosen so that, whenever the algorithm has to compute $V(x)$, all $V(x_a)$ have already been computed and $V(x)$ can be computed from equation (9). The complexity of a dynamic programming algorithm is $O(Sm)$, in which S is the number of possible states x for which we have to optimize the Bellman equation and m is the maximum number of actions in one state. For many problems, S increases exponentially with the size (called 'the curse of dimensionality'), often implying an exponential run time for dynamic programming.

Quantum speedups are known for a substantial number of classical dynamic-programming algorithms, utilizing a combination of classical dynamic programming and Grover's search. For example, the quantum algorithm for the TSP in ref. 141 first uses dynamic programming to compute the shortest paths through sets of cities that contain up to 24% of all cities and then uses Grover's search to find the shortest combination of those paths that visits all of the cities and returns to the starting point. Similar ideas can be applied to other problems, in which the classical algorithm uses an exponentially large dynamic-programming table indexed by subsets of an n-element set. Examples include graph colouring[142], the minimum Steiner tree problem[143,144], tree width[145] and single-machine scheduling[146]. The problems amenable to this approach are typically vertex-ordering problems (in which the task is to find the optimal ordering of vertices in a graph, as in the TSP) or set-partitioning problems (in which the task is to find the optimal partition of a set, as in set cover or graph colouring). Such problems lead to a dynamic programming table in the form of a Boolean hypercube (2^n entries indexed by $x \in \{0,1\}^n$), which can often be handled by computing part of the table and then using Grover's search to find the best solution, leading to an overall – usually subquadratic – speedup.

A generic dynamic programming problem can be reformulated with the dynamic programming table in the form of hypercube, with the result that, in the generic case, the quantum speedup can be at most quadratic[141]. It is an open problem whether quantum speedups can be obtained for algorithms with a dynamic programming table of a different structure.

Optimal control

Optimization with differential equations as constraints is a subject of optimal control theory. A standard finite-horizon optimal control problem takes the following form: find a control function u with values in a given set $\mathcal{U}$ solving

$$\begin{aligned} \min_{u(t)\in\mathcal{U}} \quad & \int_0^T \ell(x(t),u(t))\mathrm{d}t + h(x(T)) \\ \text{s.t.} \quad & \dot{x} = f(x,u), x(t)\in\mathbb{R}^n, \quad t\in[0,T), \end{aligned} \tag{10}$$

for some nice functions ℓ, f and h (see, for example, ref. 147). A globally optimal solution of this problem, the so-called feedback optimal-control policy $u^*(x,t)$, can be found by introducing a value function:

$$V(z,t) = \min_{u(t)\in\mathcal{U}} \left\{ \int_t^T \ell(x(s),u(s))\mathrm{d}s + h(x(T)), \dot{x} = f(x,u),\ x(t) = z \right\}. \tag{11}$$

Generalizing classical dynamic programming ideas, it can be demonstrated that V solves the following Hamilton–Jacobi–Bellman (HJB) equation backwards in time:

$$\frac{\partial V}{\partial t} + \min_{u\in\mathcal{U}}\{\ell(x,u) + f^\top(x,u)\nabla_x V(x,t)\} = 0, \tag{12}$$

subject to the terminal condition $V(x,T) = h(x)$, for all $x\in\mathbb{R}^n$. Given the solution of equation (12), the optimal-control policy in the feedback form is computed as follows:

$$u^*(t,x) = \operatorname*{argmin}_{u\in\mathcal{U}}\{\ell(x,u) + f^\top(x,u)\nabla_x V(x,t)\}. \tag{13}$$

The key problem associated with the HJB equation, continuous or discrete, is the 'curse of dimensionality'. Solving equation (12) by conventional methods of numerical analysis quickly becomes intractable even in rather modest dimensions (for example, if $x(t)\in\mathbb{R}^n$ for $n\geq 10$). In other words, a direct approach to solving the HJB equation has slim chances of succeeding classically.

In addition to optimal control, the HJB equation has many applications: from nonlinear filtering or state estimation for differential equations[148] to discrete optimization – the most relevant application in the context of this Review. For instance, the HJB equation for discrete-time dynamics can be seen as the famous Bellman equation of dynamic programming, which in turn connects HJB equations with reinforcement learning, a family of algorithms in which an agent aims to learn optimal decisions in unknown environments through the experience of taking actions and observing the rewards gained (see the Supplementary information for details).

Quantum approaches can be classified based on whether the learning agent has a standard classical access to the environment or whether the interaction itself is quantum. In the latter case, numerous types of quantum oracular access to the task environment have been explored: for example, where coherent access to a unitary encoding the transition function is possible[149], or where the environment closely mimics classical reinforcement-learning settings, allows only actions as inputs and has memory[150]. The main conceptual limitations are the requirement to coherently access the task environment, which might only be possible in very restricted or constructed cases. In some works, the focus was also on identifying the exact solutions, for which, in real-world problems, the search spaces are simply too large. The up-side of the aforementioned results is that the speedups concern the number of interactions with the environment and are not contingent on any assumptions in complexity theory.

It has been shown[151] that quadratic speedups in policy optimization are possible as long as the policies satisfy certain regularity conditions – which are, for example, satisfied for policies defined by parameterized quantum circuits. The lower bounds for this problem have been established for a more general setting[152] and they show that these quadratic improvements cannot be improved further. However, if the environments are not arbitrary but special, more substantial improvements are possible, including provable exponential speedups[153]. The contrived nature of these environments makes these results unlikely to have implications in practice.

When the access to the task environment is restricted to be classical, provable speedups are possible too. For example, it has been proved[154,155] that there exists task environments with a super-polynomial advantage for quantum reinforcement-learning algorithms (with respect to finding optimal policies and optimal q-functions), unless the discrete logarithm problem has an efficient classical solution.

Benchmarks

Benchmarking can provide valuable insights into the practical performance of optimization algorithms in situations in which purely analytical methods are insufficient. In fact, it can even be misleading to rely only on analytically assessed performance because theoretical complexity analysis typically pertains to general problem classes and considers mostly asymptotic scaling behaviour or worst-case scenarios. The theoretical worst-case bounds often fail to describe the difficulties of solving real-world applications that focus on a specific problem instance of practical use, nor do they faithfully describe how well practically relevant problems can be solved for fixed size and data. Meaningful and systematic benchmarks are therefore essential to demonstrate practical usefulness of optimization algorithms. However, conducting such benchmarking with well-defined assumptions, considering implementation, data and metrics, is a challenging task.

Here, we will highlight best practices for benchmarking and provide a set of interesting problems that enable a thorough, robust, fair and reproducible framework for comprehensive comparisons of different classical and quantum approaches. We first give an overview of related work, then present benchmark design choices such as the metrics to be evaluated and finally elaborate on various optimization problems, which are difficult for state-of-the-art solvers and might provide interesting benchmarking problems for quantum methods.

Related work

General introductions to computational benchmarking can be found in refs. 156–161. Several important collections of classical optimization benchmarks have originated from publicly hosted 'benchmarking challenges' or competitions. The aim of these challenges is to understand and improve the practical performance of algorithms for various optimization problems, particularly those that are theoretically hard. The challenges aid in evaluating realistic algorithm performance conditioned on the resources available at the time of the competition. Some of the most relevant, regularly held optimization challenges, tackling problems using classical digital computers, are the Discrete Mathematics and Theoretical Computer Science (DIMACS) implementation challenges (http://dimacs.rutgers.edu/programs/challenge/) and the Satisfiability competition[162].

Optimization competitions and libraries that are compatible with (near-term) quantum computers have not yet been widely established, but would enable streamlined benchmarking of quantum optimization methods. Previous work has focused on application-centric quantum-hardware benchmarking frameworks that employ optimization problems as an exemplary use-case[163–166]. We give an overview of selected experimental quantum optimization results run on gate-based quantum computers with more than 16 qubits in the Supplementary information – these results represent excellent starting points for comprehensive quantum optimization benchmarks. Further benchmarks that investigate different properties of quantum annealing methods – some including comparisons to classical algorithms – for optimization problems can be found in refs. 166–175. A comparison of Ising machines that checks success probabilities for finding the ground state and the corresponding time-to-solution is presented in ref. 176. Various QUBO solvers are compared in ref. 177 with respect to the solution quality and time-to-solution, which is a good baseline for an optimization benchmark.

None of these works offers holistic optimization benchmarking frameworks, but they do provide useful insights that can help to define reproducible, representative and fair benchmark libraries, which will facilitate identifying the advantages and disadvantages of various (quantum) solvers.

Design

To ensure reproducibility as well as replicability of a benchmark, it is important to clearly define the specifics of model selection, pre-processing, computational platform, target metrics and selected algorithm (provably exact, heuristic and so on) including suitable hyperparameters. Relevant algorithmic paradigms are given in Box 1, and quantum algorithms for optimization problem classes have already been presented. We now discuss the remaining levels of design choices.

Model independence. To find the optimal (or at least a sufficiently good) solution of a concrete problem, the optimization problem must be formulated in terms of a rigorous mathematical model, such as MIP or QUBO, before selecting a particular algorithm or computational platform. The model choice is usually not unique and can strongly affect the performance of algorithms: certain models and algorithms are better suited for certain problems than others. Hence, to find the overall best solution to a given problem, a model-independent benchmark must be defined, allowing all possibilities to model a certain problem. Alternatively, the aim might be to find the best algorithm or hardware platform to solve a problem given a specific model formulation, which results in a model-dependent benchmark.

Model-independent benchmarks without limitations on admissible solution strategies are crucial to demonstrate quantum advantage for optimization. Quantum advantage may stem from novel problem formulations that could not be captured on a model-dependent level. Furthermore, quantum hardware is evolving quickly, and model-independent benchmarks are better suited for comparability and compatibility between different generations of hardware, enabling leverage of new features that might not have been available in earlier generations, such as dynamic quantum circuits[178]. We expect these benchmarks to eventually show how quantum optimization algorithms are approaching, meeting and exceeding the capabilities of classical techniques for solving particular problem instances.

In the remainder of this section, we focus on the model-independent setting. However, we would like to highlight the importance of model-dependent benchmarks, for example, in comparing hardware capabilities of different gate-based quantum computers by solving a given QUBO model and in tracking progress in hardware and algorithmic development.

Pre-processing. The term 'pre-processing' or 'pre-solving' refers to a set of methods that are applied to a problem instance or data before the actual optimization algorithm is executed. The goal of pre-processing is to modify the problem in such a way that it enables the optimization algorithm to work more efficiently. Pre-processing can have a substantial impact on benchmarking results and should be chosen and reported on with care.

How pre-processing is implemented in practice depends on the data, the optimization algorithm and its mathematical formulation and on the computational platform. In fact, it is not uncommon for many problems – such as the Euclidean TSP, or Steiner tree problems in graphs – that pre-processing is able to remove 95% of the variables and, hence, substantially reduce the problem size and resource requirements[179,180]. Considering the example of QUBO, various reduction techniques to eliminate optimization variables have been suggested in the literature and are reviewed in ref. 180. For example, analytic reduction rules[181] make it possible to conditionally fix certain optimization variables, while guaranteeing that at least one optimal solution will remain in the search space. In addition to the number of optimization variables, their degree of correlation (which is determined by the sparsity of the QUBO matrix) is also of interest. In the case that certain groups of decision variables are not correlated, QUBOs can be decomposed trivially into smaller sub-QUBOs to be solved independently.

More generally, decomposition approaches can be an option to break large problems into smaller subproblems. Decomposition is well established in classical optimization but can also help quantum optimization – for example, in fitting a large problem on a smaller quantum computer. However, decomposition often introduces a trade-off between the size and number of subproblems and the overall resulting solution quality. Thus, achieving a quantum advantage in the subproblems is necessary but not sufficient for achieving a quantum advantage for the global problem. But as long as the advantage can be carried over, decomposition can be a key tool towards quantum advantage. The development of quantum-specific decomposition schemes might further help achieve this goal.

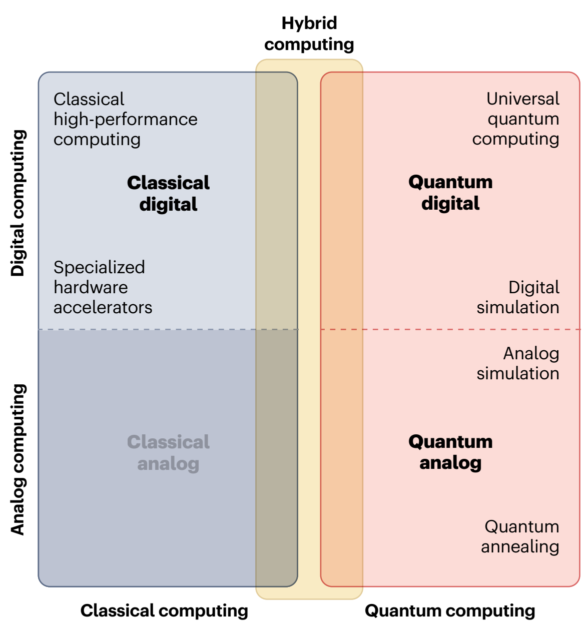

Fig. 2 | The optimization-platform ecosystem. Various hardware types are relevant for solving optimization problems – digital or analog, classical or quantum, and even hybrid computing schemes – some better suited to certain optimization problems than others. In our discussion of benchmarking, we focus on digital classical computing and on digital and analog quantum computing.

Platforms. Different platforms have different strengths and suffer from different bottlenecks, and hence some may be better suited for certain optimization problems or modelling tasks than others. Furthermore, depending on the platform, the optimization models and applicable algorithms do vary, resulting in better or worse performance for a given problem. Understanding the intrinsic properties of a hardware platform is crucial to put the results of a benchmark into perspective and to define benchmarks that allow for fair comparison between systems that can function very differently – a non-trivial task. Cross-platform benchmarking can be used to better understand the advantages and disadvantages of different platforms for different optimization problems.

We will now elaborate on the various hardware types that are relevant for optimization problems – that is, classical versus quantum, and analog versus digital. An illustration of the platform ecosystem is shown in Fig. 2.

Classical digital systems are the workhorse of today's computer technology. The basic units of information are the binary values 0 and 1, and information is processed by compositions of logical operations such as AND and NOT gates, which can be expressed in a circuit model. Digital classical computational models often rely on a combination of central processing units (CPUs), which excel at sequential and general data-processing tasks, and graphical processing units (GPUs), which specialize in parallel processing tasks.

There are several other types of processors and specialized hardware accelerators designed for specific tasks and applications, for example, tensor processing units for deep learning and field-programmable gate arrays for signal processing and embedded systems, to name just two. The combination of CPU, GPU and special-purpose processors forms the foundation for classical high-performance computation in a wide range of applications. Classical digital hardware devices are incorporated in a wide range of computing platforms, including desktop computers, computing clusters, cloud services, mobile phones and many more. All of these platforms can be used to solve optimization problems.

Special-purpose systems for optimization also exist, developed to simulate quantum phenomena such as quantum annealing. Well-known examples of this type of technology are the Fujitsu Digital Annealer[182,183], which is based on a hardware-accelerated

Markov-chain Monte Carlo approach; the Hitachi CMOS Annealing Machine[184], whose in-memory computing architecture represents an Ising model with local interactions; and the Toshiba Simulated Bifurcation Machine[185,186], which does not approximate an annealing process but instead implements a highly parallelizable approximation of Ising dynamics using classical hardware (GPU and field-programmable gate arrays), enabling computation of heuristic solutions to problems with up to millions of variables in a time frame of seconds.

In digital quantum computing[187], the basic units of information are qubits instead of bits. The information processing consists of consecutive quantum gate operations such as CNOT gates and single-qubit rotation gates, which, together, form a quantum circuit, similar to a classical logical circuit. Important bottlenecks of existing digital quantum computers are limited coherence times, which result in limited circuit depths, limited qubit numbers and limited qubit connectivity. There are various ways to quantify the performance of a digital quantum computer. An overview of best practices for executing quantum optimization algorithms on noisy digital quantum hardware is given in the Supplementary information.

Analog quantum computing hardware is a more specialized kind of quantum computer that performs continuous time evolution of the Schrödinger equation[188]. These types of computers are a natural fit for quantum simulation applications[188–196]. Additionally, the quantum annealing algorithm (discussed earlier) provides a way for these computers to be used for unconstrained optimization tasks as well[2,3,167,197,198]. Thus far, analog quantum computers have been developed for solving optimization problems in the format of Ising models[199] and special cases of the maximum independent set[200]. In contrast with digital quantum computing, which is very flexible, analog quantum computing focuses on a more limited scope of optimization tasks, compatible with analog hardware.

Metrics. Owing to the technological differences between classical and quantum systems, it is crucial to select metrics that enable a fair comparison. The main metric categories are resource cost and solution quality.

The resource cost includes any quantum or classical computational resources that are employed to solve an optimization problem, including any potential pre-processing or post-processing steps and resources that are used in parallel. Classical computing resources are typically measured with respect to factors such as used memory, required memory bandwidth and CPU specifications, including clock speed and the number of cores. Quantum computational resources can be measured in terms of the number of circuits, the number of executed gates per circuit, the number of qubits and shots per circuit execution and the quality of the qubits. Of course, the resource cost should also consider resources that are required for problem-dependent calibrations and error mitigation or, if applicable, correction. The number of qubits in a quantum programme strongly depends on the mapping of problem variables to qubits; hence, algorithmic improvements in mapping schemes can lead to a reduction in qubit requirements. Resource-cost reporting must also include the run time, that is, the duration of usage of quantum or classical computational resources.

Unfortunately, several of these resource costs are difficult to compare over different platforms and algorithms. Although, ideally, it would be good to check absolute metrics such as the total energy usage required to get a solution, that kind of information is usually difficult to access for the end-user. The overall run time is one of the most important metrics for providing an objectively fair comparison of different computational platforms – but this still strongly depends on the benchmarking setting (such as what pre-processing is used), and it is hence of utmost importance to clearly define what determines a faster algorithm. The effective end-to-end run time of an algorithm consists of the time used for pre-processing, transpilation, embedding, compilation, execution and post-processing. For existing quantum computing software, transpilation, embedding and compilation times contribute substantially. However, these are areas of active research and development and we expect the times to be reduced in the near future.

The solution quality strongly relies on the nature of the problem. First, a solution must be compatible with the problem constraints – that is, it must be feasible, which can be easily checked. Ultimately, the hope is to find the optimal solution or at least a solution that is ε, which is close to the optimal solution, but note that the evaluation of optimality might be resource-intensive or even intractable. Given a feasible solution, the optimality gap can be identified via bounds given by continuous relaxations[201] or dual representations[202]. Furthermore, in certain applications, a set of diverse solutions, of comparable (good) quality, might be of interest: for instance, in the case that a complex real-world optimization problem has to be represented by a simplified model. Given a diverse set of solutions, a decision-maker may pick the one that best matches (possibly informal) constraints and preferences that could not be directly added to the optimization problem. Other examples are multi-objective optimization (Supplementary information) and accelerating branching strategies in MIP solvers. Solution diversity is difficult to quantify in general as its utility depends mostly on the context.

To provide a holistic quantum optimization benchmark, metrics that describe algorithm and solution properties should be measured. Depending on the problem, model, algorithm and execution platform, the respective metrics should be chosen carefully and reported transparently.

Problems

We will now present a set of model-independent problems that show promise for establishing standardized quantum optimization benchmarks. The problem instances chosen may be crafted, random or real-world problems.

Crafted instances, designed to represent challenging problems, are most likely the best way to find problem instances that are hard to solve by classical methods. However, there is a risk that there is also a classical method (or one could be developed) that takes advantage of the special problem structure or that the instances are of no practical value. Random instances are different because they lack structure. Depending on the problem, this can make them particularly easy or very difficult to solve, and how meaningful these instances are can be debated. Real-world instances correspond in many practically relevant cases to MIPs that can be solved up to sufficient accuracy using existing solvers. However, instances that are relevant and difficult to solve with existing methods are hard to find in a setting that fits the dimensionality and density limitations of current quantum hardware. In fact, these problems are generally hard to find because open problems of industrial relevance are rarely published or may be ill-defined. This highlights a common selection bias: we rarely model practically relevant problems in a way we know we cannot solve.

The problems chosen should be sufficiently hard for state-of-the-art classical solvers to leave a margin for quantum methods, but should also be within an intermediate size range in which current or near-term quantum technologies can be effectively deployed. Ideal problem instances are those for which existing methods are unable to provide provably optimal or feasible solutions. Furthermore, it is crucial that the problem can be formulated as a quantum-compatible model, that is, with sufficiently few variables and qubits, limited connection density, relatively small coefficients and so on. In other words, a sweet spot for current quantum systems is sought, which enables the investigation of quantum advantage compared with suitably chosen digital algorithms. It can also be valuable for tracking the progress in algorithmic and hardware development to identify a family of problems that are similar in their nature but differ in their difficulty.

The following are binary problems that are promising candidates for quantum optimization benchmarking problems (additional details can be found in the Supplementary information). These problems are random or crafted instances and relate to only a few practical applications. Nevertheless, they offer a good testbed for algorithm development and progress tracking.

Maximum independent set. This is a classic NPO-hard problem and there are known instances (https://oeis.org/A265032/a265032.html) that are hard to solve to proven optimality using existing methods, starting at problem sizes of several hundred variables. The allure of this problem class for quantum optimization benchmarking is that it is well suited for translation into relatively sparse QUBOs[203–206].

Knapsack or market share. The goal of a (multidimensional) knapsack problem (or market share) is to find the maximal set of items that can be added according to a set of given weights, such that the resulting sum is lower or equal to a given knapsack constraint. For carefully chosen coefficients, these problems can be difficult to solve using classical methods to proven optimality[15,207,208]. We believe that this problem class is a good candidate for quantum optimization benchmarking because there is clear evidence that it is already challenging for classical digital systems to find (approximate) solutions – even for small system sizes in the multidimensional case.

Low autocorrelation binary sequences. This is a difficult binary nonlinear optimization problem[13], with the goal of finding, using binary variables, a sequence S of length k that minimizes a given energy function. There is only one particular instance per size N; hence, the frontier of what is possible is always well defined. Given the spin-like binary variables, it might be possible to find a model formulation that could provide a natural fit for a quantum platform, although the quartic objective function creates challenges in doing so[209].

Quadratic assignment problem. This problem[210] has remained one of the great challenges in combinatorial optimization. It is still considered a computationally nontrivial task to solve to optimality, even for modestly sized problems ($n \approx 30$ facilities and locations). Modelling this problem as a QUBO results in a dense formulation with n^2 variables. The long and unsuccessful search for improvements in classical methods indicates that there might be room here for quantum advantage.

Sports timetabling problems. These problems describe scheduling problems for sports leagues or tournaments and are generated in a way that ensures at least one feasible solution exists per instance[211]. Interestingly, for several medium-sized instances, no known (and investigated) method has been able to find even a single feasible solution[212,213]. What is particularly interesting about these problems is that instances can be generated at variable size and difficulty to track progress, but it is known that sufficiently large problems are difficult for state-of-the-art methods.

Spin glasses. The mapping of a spin-glass problem onto a QUBO is trivial and leads to relatively sparse models. Classical solvers often perform well on spin glasses defined on regular lattices; however, the possibility of digital quantum optimizers having better performance still exists – possibly on non-regular lattice structures[214]. Hence, spin-glass problems are interesting candidates for quantum benchmarks, especially in a cross-platform setting. Moreover, spin-glass problems have properties that make them provably average-case hard for various families of classical algorithms[21]. It is an open question whether quantum algorithms can do better in these average-case settings, and thus, superpolynomial speedups for spin-glass problems are still possible.

Classical methods that can be used to approach all of these problems include branch-and-cut-based integer (linear, nonlinear or semi-definite) programming[215], pseudo-Boolean optimization[216], satisfiability solving[162,217], constraint programming[218] and several other general techniques[219] as well as a multitude of specific heuristic approaches.

The path towards quantum advantage

Classical algorithms can be developed both theoretically and empirically. Algorithms that work well empirically may only later be fully understood theoretically. By contrast, quantum algorithms are primarily developed theoretically owing to limited hardware availability. However, this is changing. The rise of gate-based quantum computers with hundreds of qubits enables complex circuits beyond exact classical simulability[220,221]. To develop and improve quantum optimization algorithms, in addition to continued technological progress, it is crucial to combine theoretical and empirical research – similar to classical algorithms. In this context, we believe that there are three key directions to advance on the path towards quantum advantage.

First, real-world optimization problems should be identified that are promising candidates to demonstrate quantum advantage. This is challenging owing to the many open questions about the future performance of quantum optimization algorithms, their heuristic nature and the overhead introduced by quantum error correction. Additionally, these problems must be truly challenging for classical solvers; if classical solvers are sufficient, there is no margin for quantum advantage. A formal understanding of the actual goal of application is required. This may sound trivial, but practical challenges can arise when trying to formally define objective functions, constraints or preferences in a rigorous mathematical way, or when trying to get the required data.

Furthermore, as mentioned earlier, problems are typically modelled to be solvable, often involving simplifications such as linearization. Although the impact of these simplifications can be negligible in some cases, it might be notable in terms of practical impact or solvability in others. Challenging and relevant problems must be rigorously defined, ideally in a model-independent way, and necessary data should be available to construct instances using different modelling approaches.

Once a concrete problem instance is identified, it should be compared with the state-of-the-art classical algorithms. Two illustrative applications (discussed further in the Supplementary information) are financial asset allocation and the sustainable energy transition:

although these may not be the first domains to benefit from quantum advantage in optimization, they are well suited to illustrate different levels of modelling complexity, data requirements and the scale needed for real-world applications. The resulting problems, including all introduced features, can be difficult to solve classically. Such a playground to construct real-world motivated benchmarking problems with increasingly complex features is essential to test algorithms and track progress from toy examples towards practically relevant problems.

The second direction involves finding application-agnostic (possibly hand-crafted) optimization problem instances and the corresponding (heuristic) quantum algorithms that can outperform the best classical algorithms. If (artificial) instances are found where quantum algorithms excel, it might be then possible to discover related applications or understand the source of the advantage and extend it to broader classes of problems or algorithms. These instances may even motivate completely new algorithmic primitives.

An example in which quantum algorithms might achieve a provably exponential speedup over the best-known classical algorithms is optimization problems, which are reductions (for example, factoring problems[222,223]). However, such examples are unlikely to occur in practice other than by encoding problems that are in NP $\cap$ BQP. Here, we suggest an empirical approach to identify more generally applicable quantum heuristics and less formal problem structures in which these heuristics have an advantage in certain instances.

For the third direction, theoretical understanding of the potential and the limitations of quantum optimization algorithms should be enhanced, and new algorithms developed that ideally achieve more than a quadratic speedup. This includes research on exact algorithms, approximations and heuristics, as well as answering more general questions such as about the potential of quantum optimization in the average case in contrast to the worst case. Theoretical insights help to identify promising directions for empirical testing and to explore scenarios that cannot be tested empirically yet. For instance, to enable practical advantages, there should be a move away from relying on QRAM for designing quantum optimization algorithms with rigorous quantum speedups: although the QRAM model has led to elegant results, its physical realization remains a distant possibility at best. Furthermore, finding quantum speedups beyond quadratic is crucial for practical quantum advantage[224]. New theories can inspire new algorithms that address this difficult task, unlock new problem classes and create opportunities to find problem instances that may benefit from quantum optimization.

To conclude, use of quantum hardware is already critical in accelerating progress towards quantum advantage. Simultaneously, state-of-the-art classical optimization must be leveraged to identify where better solutions are needed. Empirical studies do not replace theoretical research, but rather they complement each other.

Outlook

We have provided a comprehensive overview of the potential, the challenges and the emerging research areas in quantum optimization. We observed that although complexity theory is useful as a guide to provable performance guarantees, it may not always be useful for finding practical quantum advantage. This underscores the need to develop and analyse quantum heuristics to augment theoretical research. We have highlighted key problem classes in optimization, reviewed existing algorithms and suggested new research directions. We have also discussed the importance of benchmarking for identifying quantum advantage.

As quantum hardware evolves, a new era is beginning in which progress in quantum algorithms comes from both theoretical analysis and empirical methods. There is a strong need to continue discovering new theories and algorithms with provable performance guarantees, but there must also be increased development of quantum heuristics and benchmarking of their performance on real quantum computers. The improving quantum capabilities unlock opportunities to further advance quantum optimization (including the co-design of quantum algorithms that account for hardware limitations, discussed in the Supplementary information). Even if there is no immediate practical quantum advantage, the intuition gained from running experiments on real hardware will at least enable quick validation of proposals and may also inspire new theoretical research.

In addition to technical advances in quantum optimization, there is also the question of responsible use of this new technology. Future research should be done with the awareness that optimization use cases occur in a multitude of contexts – some with positive societal impact, others with nefarious intent. We hope that open scientific discussions will encourage the advance of quantum optimization and we advocate responsible use of the insights presented in this article. Where applications of quantum optimization algorithms are chosen and funded, we encourage the prioritization of those with positive social impact.

Quantum optimization holds vast potential for various applications, but challenges remain in demonstrating practical quantum advantage. When a tangible quantum advantage is demonstrated, we expect that quantum optimization will rapidly influence many domains, given the widespread relevance of optimization.

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Author contributions

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Competing interests

The authors declare no competing interests.

Additional information

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